Optimizing Quantum Teleportation Protocols

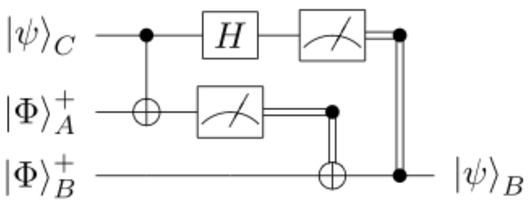
Yulie Arad[1,2], Mayank Bhatia[3], Felix Leditzky[2]
[1]Department of Computer Science, UIUC [2]Department of Mathematics, UIUC [3]Department of Electrical and Computer Engineering, UIUC

MOTIVATION

- Quantum Teleportation is a way to transmit quantum information using entangled resources.
- The No-Cloning Theorem states that it is impossible to duplicate any arbitrary state.
- The standard teleportation protocol [1] for communication works theoretically, but practically, there are many errors, also known as noise.
- Different resources need different protocols to be optimal.

Quantum Teleportation

- **Quantum Teleportation** is used to transfer quantum bits (qubits) between two users. The standard teleportation process requires an entangled pair of qubits and 2 classical bits. This protocol can be generalized for qutrits and qudits of higher dimensions.



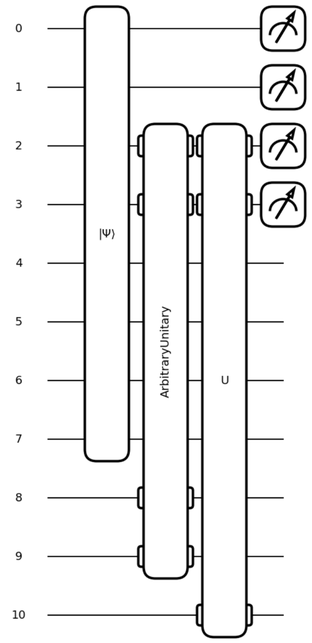
Source: "Quantum teleportation" *Wikipedia*

Reduction Criterion

- **Reduction Criterion** is a criterion that can be used to detect entanglement [2].
- For the states considered in this work, the reduction criterion is satisfied if the largest eigenvalue is $\leq \frac{1}{d}$, where d is the local dimension.
- If a state violates this criterion, it can be used in a teleportation protocol that is better than any classical protocol [3].

METHOD

- Begin with an entangled qutrit Werner state
- The goal is to perform a local operation on Bob's side, such that the resulting state violates the reduction criterion.
- Using a variational quantum optimization ansatz, we try to find such a local operation for each value of $p < \frac{1}{2}$
- Variational Quantum Optimization (VQO) is the optimization of the internal parameters of each gate in a quantum circuit. In the figure 2, it is the "ArbitraryUnitary".
- The operation must be noisy to change the eigenvalues.
- Thus, the unitary acts on wires 2, 3, 8, and 9, and we discard wires 8 and 9.

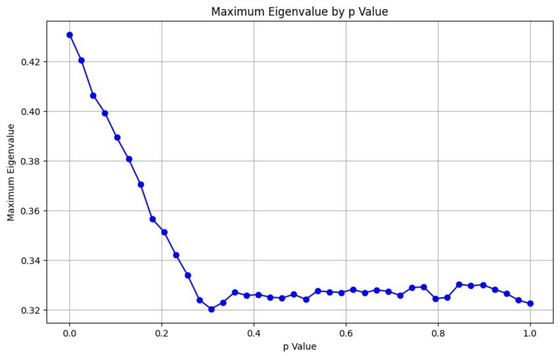


Circuit

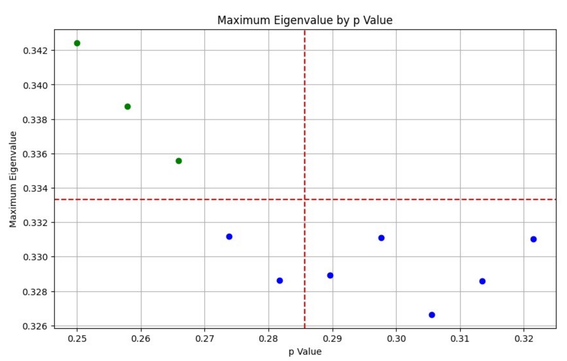
- The circuit starts by preparing the Werner state W_p shared by Alice (wires 0,1) and Bob (wires 2,3), wires 4-7 are auxiliary wires.
- We embed each qutrit into 2 qubits using the following map:
$$|00\rangle \to |0\rangle, \qquad |01\rangle \to |1\rangle, \qquad |10\rangle \to |2\rangle$$
- The circuit applies the "ArbitraryUnitary" explained above.
- When applying the operations, there is a possibility of leaving the code space. In this case, we apply U (with auxiliary wire 10) to return the state back to the code space.

Results

- We used PennyLane [4] and qNetVO [5] to simulate the circuit and get these results.



Comparison of p Value and Maximum Eigenvalue with $0 \leq p \leq 1$



Comparison of p Value and Maximum Eigenvalue with $0.25 \leq p \leq 0.32$

- For values $p \leq \frac{1}{2}$, the VQO ansatz found a quantum operation such that the resulting state violates the reduction criterion.
- This means, that for these states there are quantum teleportation protocols that are better than any classical protocol.

Outlook

- Perform the optimization for larger circuits with more auxiliary qubits.
- Test different families of quantum states and higher dimensional Werner states.
- Use a quantum algorithm for checking the reduction criterion.

Werner States

- The Werner states family is a special type of symmetric quantum state, parametrized by a single real value.
- For qutrits it can be defined using the following matrix:

$$W_p = \begin{bmatrix} \frac{p}{6} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \frac{2-p}{12} & \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \frac{2-p}{12} & \cdot & \cdot & \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot & \cdot \\ \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot & \frac{2-p}{12} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \frac{p}{6} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \frac{2-p}{12} & \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot \\ \cdot & \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot & \cdot & \cdot & \frac{2-p}{12} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \frac{-1}{6}+\frac{p}{4} & \cdot & \frac{2-p}{12} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \frac{p}{6} \end{bmatrix}$$

- The 2 distinct eigenvalues of a qutrit Werner state W_p are equal to $\frac{1-p}{3}$ and $\frac{p}{6}$
- Therefore, Werner states always pass the reduction criterion, but are entangled for $p < \frac{1}{2}$

REFERENCES

[1] Charles H. Bennett, Gilles Brassard, Claude Crépeau, Richard Jozsa, Asher Peres, and William K. Wootters Physical Review Letters 70, 1895 (1993)
[2] Michal Horodecki and Pawel Horodecki, Physical Review A 59, 4206 (1999)
[3] Eric Chitambar and Felix Leditzky, IEEE Transactions on Information Theory (2023)
[4] V. Bergholm et al., arXiv preprint arXiv:1811.04968 [quant-ph] (2018)
[5] Brian Doolittle and T. Bromley, https://github.com/ChitambarLab/qNetVO

ACKNOWLEDGEMENTS

I would like to thank Professor Eric Chitambar, the Leditzky Group and the Chitambar Group for their invaluable support throughout this past year, as well as the IBM-Illinois Discovery Accelerator Institute for this opportunity.